Malav Patel

mpatel636@gatech.edu | malav-p.github.io | github.com/Malav-P

Skills

Languages: Python, C, C++, TypeScript, CUDA, Julia, MATLAB, SQL
Frameworks: PyTorch, JAX, sklearn, HuggingFace, Docker, Kubernetes, AWS, FastAPI, PostgreSQL, Redis, CI/CD
Research Topics: Diffusion, NLP, Filtering, Optimization, Control, MPPI, HPC

Experience

AI/ML Research Intern, NASA Langley Research Center – Hampton, VA — June 2026 – August 2026

- Engineering an internal coding agent for NASA, interfacing directly with secure in-house large language models for mission-critical software workflows.

Research Engineer, Georgia Institute of Technology – Atlanta, GA — August 2022 – Present

- Applying gradient-based optimization, integer programming, and filtering to optimize placement and tasking of satellite constellations in cislunar space.

Software Engineer, AthenaLabs – Atlanta, GA — January 2026 – May 2026

- Worked on a backend agnostic execution and orchestration framework for computational experiments at scale, allowing users to run experiments locally via Docker or on large K8s/Ray clusters.

AI Trainer, Data Annotation – Atlanta, GA — June 2023 – January 2025

- Engineering difficult prompts and problems to challenge SOTA multi-modal models across english, mathematics, and science tasks. Effected 1.3% improvement in average model accuracy.

Machine Learning Engineer, Startup – Atlanta, GA — August 2025 – December 2025

- Fine-tuned Vicuna for text/image-conditioned floorplan generation, reducing training memory by 70% using LoRA and QLoRA to enable consumer-GPU deployment.
- Direct Preference Optimization to align model on layout preferences, yielding 34% improvement in quality.

Projects

On Device GPT Inference — github.com/Malav-P/staticgrad

- Built a library in C++ for running training and inference of GPT2. Optimized inference on Apple Apple M1 using Accelerate GEMM and custom attention kernels, achieving 25 ms/token latency without compiler optimizations.

Open SoundStream — malav-p.github.io/soundstream

- Established an open source implementation of SoundStream, a neural audio codec underpinning SOTA generative audio models in 2 weeks.
- Trained on a single HPC NVIDIA L40S GPU node in less than 15 hours. Released checkpoints for other users.

ModernBERT for Patent Classification — github.com/Malav-P/modernpatentBERT

- Trained modernBERT on the patent classification task with >2x speedup in training throughput achieving SOTA precision and F1 scores on a domain test set.
- Hosted an open repository on Huggingface of 3 million patents for future work on the classification task.

CNNs in C++ — github.com/Malav-P/CNN

- Built a C++ header-only library for constructing, training, and running convolutional neural networks.
- Achieved hand-written digit recognition at 60 Hz on Apple M1 with a CNN trained on MNIST.

Education

Georgia Institute of Technology – PhD in Aerospace Engineering — December 2026
Georgia Institute of Technology – MS in Computer Science — December 2026
Northwestern University – BS in Physics, Mechanical Engineering, Integrated Sciences — June 2022